Supplementary Material: A single-parameter geometric model of contextual overlap in the Gender–Science IAT

Supplementary Methods

Binning and normalization

Within each participant and critical block, trials were ordered by presentation sequence and mapped to a normalized position variable $x \in [0, 1]$. Reaction times were aggregated into quantile bins with $N_{\text{bins}} \in \{4, 6, 8\}$. Participants with fewer than three populated bins were excluded so that the cosine–sine basis remained identifiable.

For predictive validation, standardization was performed in a leak-free manner: z-scores were computed from training bins only and then applied to held-out bins. For the estimation of the pooled interference angle θ, all bins were used for standardization to maintain consistency with the Gaussian observation model used in profiling.

Numerical stability

Closed-form ordinary least squares (OLS) estimation was used throughout. To stabilize matrix inversion in near-degenerate cases, a ridge penalty of 10^{-5} was added to $X^\top X$. Results were insensitive to the precise value of this penalty within several orders of magnitude.

Supplementary Results

Robustness of the pooled posterior to binning resolution

To assess sensitivity to temporal discretization, the full profiling pipeline was repeated using 4, 6, and 8 quantile bins per participant. The inferred pooled posterior remained concentrated in the low-angle regime across all three resolutions, although the posterior mode shifted modestly with the number of bins—from 22.75° (4 bins) to 17.25° (6 bins) to 15.50° (8 bins). All three posteriors are sharply concentrated (SD $< 1°$). This pattern supports robustness of the low-angle solution while also showing that it is not numerically invariant to discretization.

Table 1: Pooled profile-posterior summary by binning resolution ($N_{\text{bins}} \in \{4, 6, 8\}$). Mean, SD, MAP, and 94% HDI are shown alongside posterior entropy (nats) and residual $\hat{\sigma}$. All posteriors are sharply concentrated.

Bins	Mean θ	SD	MAP	HDI low	HDI high	Posterior entropy	$\hat{\sigma}$ at MAP
4	22.901	0.623	22.750	21.750	24.000	2.330	0.443
6	17.292	0.483	17.250	16.500	18.250	2.077	0.630
8	15.566	0.423	15.500	14.750	16.250	1.944	0.697

Population-level bootstrap stability

To assess whether the pooled θ estimate is stable across random participant draws, a bootstrap subsampling analysis was performed. One hundred independent subsamples of $N = 30{,}000$ participants each were drawn from the full dataset (with replacement); for each subsample the complete grid-profiling procedure was repeated. The posterior MAP equalled 17° on every replicate, yielding a bootstrap SD of 0.00° and a 95% percentile interval of $[17°, 17°]$. This perfect stability, achieved on subsamples comprising only 21% of the full dataset, demonstrates that the $\approx 17°$–$18°$ result is not a feature of any particular participant composition and that the concentration of the pooled posterior reflects an aggregate signal robustly present throughout the sample.

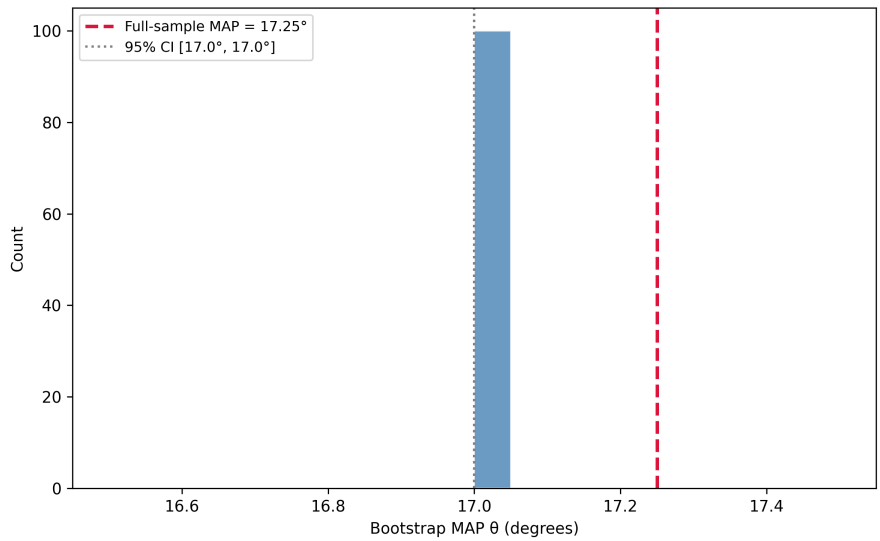

Figure 1: Bootstrap stability of the pooled θ estimate. Each of 100 replicates resamples 30,000 participants and recomputes the pooled MAP. The bootstrap distribution is effectively degenerate around the empirical solution, indicating extremely high stability of the population-level estimate at large N.

Expanded simulation-based recovery

Simulation-based recovery was conducted using 100 independent replicates for each of 11 generating angles spanning 5°–120°, with $N = 5{,}000$ simulated participant curves per replicate. Recovery was performed on the full 0°–180° profiling grid (0.25-degree steps).

Recovery is near-unbiased and yields coverage $\geq 91\%$ for $\theta_{\text{true}} \geq 40°$. For small angles ($\theta_{\text{true}} \leq 25°$), recovery is substantially biased toward the range 38°–44°, and 94% HDI coverage drops to 0%–50%. This asymmetry is geometrically expected: for $\theta \approx 17°$, the cosinusoidal argument $\theta \cdot x$ spans only a narrow range over the trial sequence, producing a nearly flat temporal modulation difficult to distinguish from additive trends at $N = 5{,}000$.

The poor small-angle recovery under $N = 5{,}000$ does not contradict the population-level estimate. At the empirically observed scale ($N = 141{,}329$), the profile posterior integrates signal across a far larger number of observations. The bootstrap analysis above—in which 30,000-participant subsamples consistently recover 17°—directly demonstrates that population-level identification is robust even though participant-level estimation remains

Table 2: Expanded simulation-based recovery (100 replicates per generating angle, $N = 5{,}000$ curves each, full 0°–180° profiling grid).

True θ	Replicates	Mean MAP	Bias	RMSE	Med. SD	94% cov.	Med. HDI
5.000	100	41.743	36.743	40.930	14.394	0.000	50.125
10.000	100	39.763	29.763	33.412	12.658	0.000	44.500
15.000	100	42.363	27.363	31.090	12.451	0.000	44.500
17.250	100	43.173	25.923	29.044	11.810	0.000	41.875
20.000	100	41.613	21.613	24.666	10.752	0.010	38.625
25.000	100	42.030	17.030	19.739	8.620	0.500	31.375
30.000	100	39.915	9.915	11.987	6.995	0.800	25.625
40.000	100	42.610	2.610	5.964	5.172	0.910	19.250
60.000	100	59.948	-0.053	2.524	2.356	0.910	8.750
90.000	100	89.988	-0.013	0.962	0.885	0.940	3.250
120.000	100	119.990	-0.010	0.494	0.438	0.950	1.500

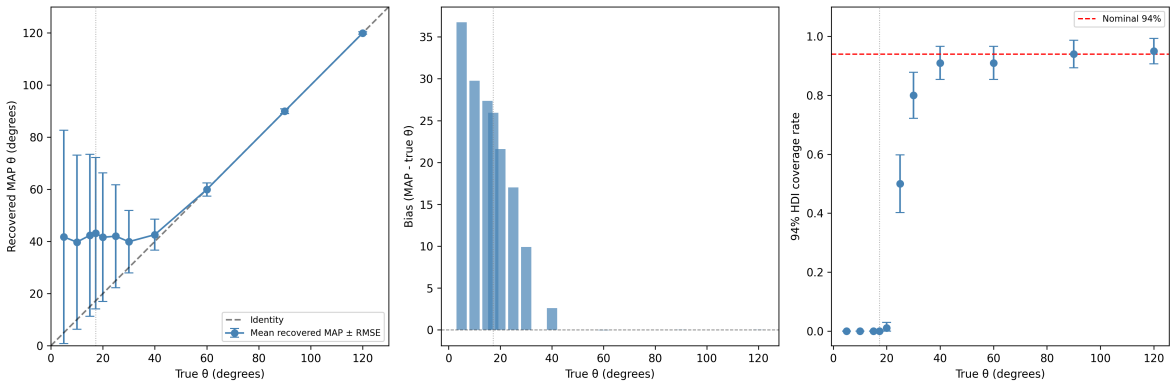

Figure 2: Simulation-based recovery across generating angles. Left: identity plot (recovered MAP ± RMSE). Centre: bias vs. generating angle. Right: 94% HDI coverage vs. generating angle. Recovery is accurate for $\theta \geq 40°$ but biased toward 38°–44° for small angles (including the empirical estimate $\approx 17°$).

unreliable in this parameter regime.

Recovery as a function of sample size

To clarify why the empirical low-angle estimate is stable despite poor recovery at small sample size, recovery was examined as a function of N. Small and moderate samples ($N \lesssim 10{,}000$) remain biased upward with 0%–50% coverage, whereas the empirical dataset falls in a regime where the recovered MAP aligns with the observed estimate and the HDI covers the target angle.

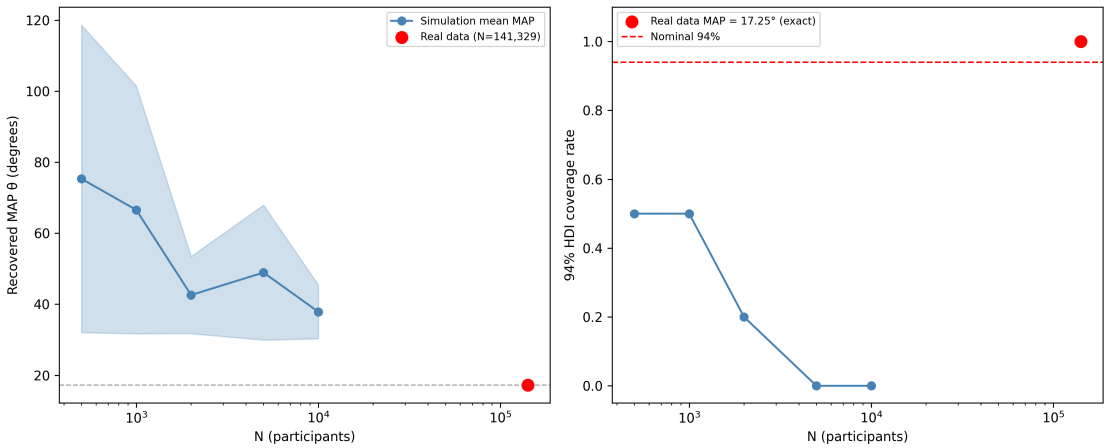

Figure 3: Population-level recovery as a function of sample size for the low-angle regime near the empirical solution. The empirical dataset at $N = 141{,}329$ lies in the regime where the recovered MAP coincides with the observed solution and the HDI covers the target.

Objective comparison: profiling posterior versus hold-out predictive criterion

Posterior density, normalized RSS, hold-out RMSE, and log-predictive-score were compared on a common θ grid for each of the three binning conditions. The hold-out RMSE is minimized at the boundary $\theta = 0°$ for all three bin counts, whereas the profile posterior peaks in the low-angle interior. This divergence reflects a mismatch between objective functions, not a contradiction in the results. Hold-out RMSE is a local predictive criterion that rewards

short-horizon interpolation of the last bins after participant-wise refitting. The profiled interference angle targets a global structural parameter—the pooled geometric relation that organizes contextual overlap across curves. These two criteria are not aligned, and the figure makes the distinction explicit.

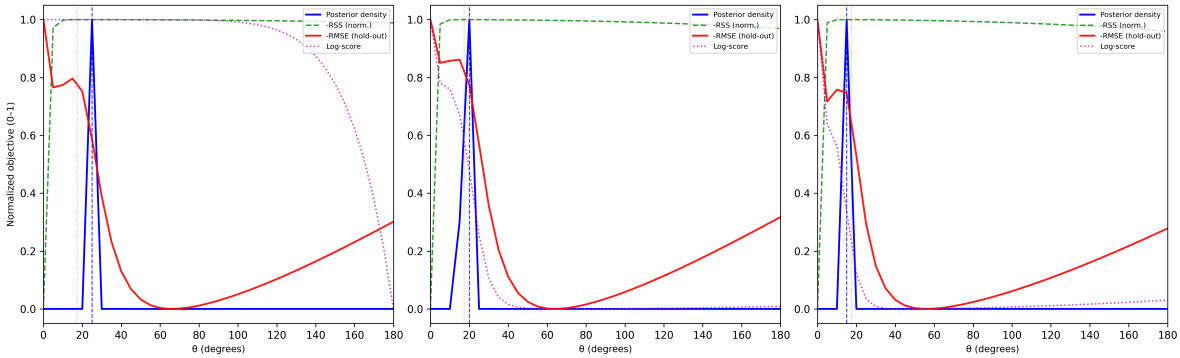

Figure 4: Population-level objectives evaluated on the same θ grid for 4, 6, and 8 bins. The posterior density peaks in the low-angle interior, whereas the hold-out RMSE optimum occurs at $\theta = 0°$ in all three binning resolutions.

Additional model comparison details

Predictive performance was evaluated using participant-wise temporal hold-out, in which the highest-$x \approx 20\%$ of bins were reserved for testing. Root mean squared error (RMSE) was computed on held-out bins after leak-free standardization. At 6 bins, differences in RMSE between all models are small in absolute magnitude: the interference model (RMSE = 21.919) and the polynomial baseline (RMSE = 21.924) are effectively tied, while cosine-only, exponential, and power-law boundary cases achieve RMSE values near 21.896–21.899. The full interference model does not achieve the lowest hold-out error, consistent with the objective-mismatch pattern described above.

Table 3: Hold-out RMSE by model and binning resolution (6 bins unless noted).

Model	Best parameter	$\text{RMSE}_{\text{test}}$	Mean log $\text{score}_{\text{test}}$	σ_{train}	Boundary hit
Interference	124.20	21.919	-21524.695	0.1056	No
Cosine only	180.00	21.899	-7212.871	0.1823	Yes
Sine only	180.00	21.897	-9632.215	0.1578	Yes
Polynomial (2nd order)	—	21.924	-21602.217	0.1055	No
Exponential	10.10	21.896	-9905.413	0.1556	No
Power law	0.05	21.897	-9259.726	0.1609	Yes

Null-geometry controls

Within-curve permutation null (500 replicates)

To test whether the recovered contextual overlap angle reflects genuine temporal structure rather than smoothness or binning artefacts, bin order was randomly permuted within participant curves and the profiling procedure was repeated. Five hundred permutation replicates were generated. Under permutation, null posterior MAPs scattered broadly (mean $= 109.65°$, SD $= 67.79°$, covering the full $[0°, 180°]$ interval), with mean posterior SD averaging 7.72°—16 times wider than the observed SD of 0.48°. The observed concentration near 17° is therefore statistically unusual under the null that temporal ordering carries no information.

Quadratic null (300 replicates)

Synthetic data were generated from the empirical polynomial (quadratic) coefficient distribution and subjected to the same profiling procedure. Three hundred replicates ($N = 5{,}000$ per replicate) were assessed on the full 0°–180° grid. The quadratic null produced MAP estimates centred near 42° (mean = 42.25°, SD = 13.59°, mean HDI width 41.70°), substantially broader and higher in entropy (mean = 5.20 nats) than the observed posterior (HDI width 1.75°, entropy 2.08 nats). Additive polynomial trends do not produce the sharp low-angle concentration present in the real data.

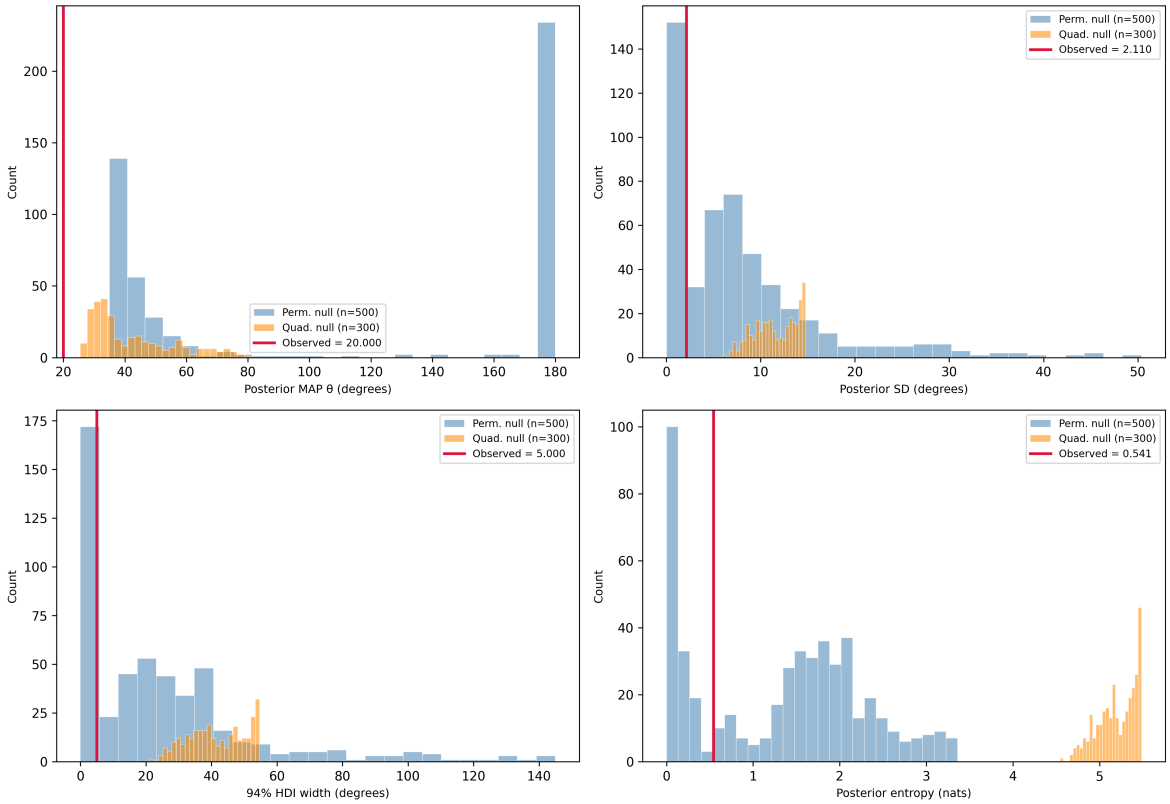

Figure 5: Null posterior diagnostics versus the observed fit. The observed posterior (crimson vertical lines) is substantially sharper and lower-entropy than either the within-curve permutation null (500 replicates, blue) or the quadratic additive null (300 replicates, orange).

Residual structure and held-out distributional fit

Near-tied RMSE on participant-level temporal hold-out does not imply equivalent model structure. On block-curve hold-out data, the interference and quadratic model RMSEs are nearly indistinguishable (57.218 vs. 57.235), but their residual maps differ systematically. The quadratic model leaves stronger negative drift in the last held-out bin, particularly in the higher-angle blocks 4 and 7 (mean absolute residual in the final held-out bin: 0.476 vs. 0.381 for the interference fit). The magnitude of quadratic residuals also correlates more strongly with the blockwise pooled angle ($r = 0.240$) than the interference residuals do ($r = 0.130$).

Considering the full held-out distribution, rather than pointwise RMSE, further discriminates the models. Across the eight held-out block/bin cells, the interference fit yields lower Wasserstein distance, lower quantile error, and lower interquartile-range error in every cell (cell means: 1.022 vs. 1.323, 0.956 vs. 1.283, and 1.524 vs. 2.070, respectively). The additive model can match local pointwise error while still misrepresenting the shape of the held-out response distribution.

Matched public Sexuality IAT: extended results

To complement the six-bin matched-domain comparison reported in the main text, the full profiling pipeline was repeated at 4 and 8 bins for both the year-matched 2019 Gender–Science and Sexuality archives. Table 4 gives the posterior summaries for each domain×resolution combination; Table 5 reports the six-bin posterior contrast.

Across all three resolutions the two domains remain in the same low-angle regime, and the same downward shift of the MAP with finer binning observed in the main corpus is reproduced in both matched archives. The posterior contrast (Sexuality minus Gender–Science) has 94% HDI $[0.50°, 3.00°]$ with posterior probability 0.991 of exceeding zero.

Table 6 extends the predictive comparison to the Sexuality domain. The interference model and quadratic baseline are again nearly tied on temporal hold-out RMSE (25.015 vs.

25.019), while cosine-only, sine-only, exponential, and power-law variants achieve marginally lower RMSE (24.997–25.000). This local-prediction tie is therefore not specific to the Gender–Science IAT.

Table 4: Matched 2019 public-domain pooled θ summaries by binning resolution.

domain	bins	n_curves	$theta_mean$	$theta_sd$	$theta_map$	hdi_low	hdi_high	$posterior_entropy$	rss_at_map
Gender-Science	4	138957	22.901	0.628	22.750	21.750	24.000	2.338	108860.878
Gender-Science	6	138955	17.294	0.488	17.250	16.500	18.250	2.086	330497.343
Gender-Science	8	138955	15.565	0.427	15.500	14.750	16.250	1.953	539470.098
Sexuality	4	207367	25.658	0.710	25.500	24.500	27.000	2.462	163468.302
Sexuality	6	207366	19.119	0.532	19.000	18.250	20.000	2.172	503741.487
Sexuality	8	207365	17.049	0.456	17.000	16.250	17.750	2.018	826124.504

Table 5: Posterior contrast (Sexuality minus Gender–Science) at six bins.

contrast	$delta_mean$	$delta_sd$	$delta_map$	hdi_low	hdi_high	$prob_gt_zero$
Sexuality - Gender-Science	1.825	0.722	1.750	0.500	3.000	0.991

Table 6: Leak-free temporal hold-out model comparison for both matched 2019 public domains (six bins).

domain	year	bins	model	$best_param$	$train_rss$	$rmse_test$	$mean_logscore_test$	$sigma_train$	n_curves	n_test_obs	$boundary_hit$
Gender-Science	2019	6	interference	124.3000	6213.0978	22.0670	-21818.2249	0.1056	138955	277910	False
Gender-Science	2019	6	cos_only	180.0000	18459.1189	22.0477	-7317.7371	0.1822	138955	277910	True
Gender-Science	2019	6	sin_only	180.0000	13822.3879	22.0453	-9770.9012	0.1577	138955	277910	True
Gender-Science	2019	6	poly2	NaN	6230.4583	22.0725	-21896.5903	0.1055	138955	277910	False
Gender-Science	2019	6	exp	10.1000	13442.1618	22.0443	-10046.7433	0.1555	138955	277910	False
Gender-Science	2019	6	power	0.0500	14436.1647	22.0454	-9393.3915	0.1608	138955	277910	True
Sexuality	2019	6	interference	119.8000	10741.8804	25.0148	-24196.6325	0.1137	207366	414732	False
Sexuality	2019	6	cos_only	180.0000	27003.5082	24.9999	-9598.2930	0.1804	207366	414732	True
Sexuality	2019	6	sin_only	180.0000	22236.5202	24.9970	-11653.6571	0.1637	207366	414732	True
Sexuality	2019	6	poly2	NaN	10757.3621	25.0192	-24275.4904	0.1135	207366	414732	False
Sexuality	2019	6	exp	8.1500	21885.1888	24.9966	-11840.5722	0.1624	207366	414732	False
Sexuality	2019	6	power	0.0700	22587.7528	24.9977	-11495.3729	0.1649	207366	414732	False

External leverage: D-like quintiles and independent life-satisfaction IAT

D-like congruency quintiles

To test whether the recovered angle is reducible to a conventional IAT effect-size summary, the Gender–Science corpus was stratified into quintiles of a D-like congruency score (difference between incongruent and congruent mean latencies divided by participant SD). Mean

D-like values vary substantially across quintiles (from -0.684 in the lowest to 0.788 in the highest), yet the subgroup posteriors remain tightly clustered in the same low-angle regime. Quintile MAP estimates range only from 17° to 18°, with HDIs spanning roughly 15°–20°. The angular summary is not captured by the conventional congruency metric.

Independent life-satisfaction IAT

An independent 2024 life-satisfaction IAT (Testable protocol, five phases) was analyzed as an out-of-sample contrast. The raw archive contained 187 per-participant CSV files; after excluding repeated names (retaining first attempt) and participants with $> 10\%$ of raw attempts faster than 300 ms, and collapsing correction loops, 180 valid participants were retained. Profiling was restricted to the critical combined phases (phases 3 and 5).

The resulting posterior concentrates at the near-orthogonal boundary: posterior mean 177.48°, SD 2.68°, MAP 180.0°, 94% HDI $[172.75°, 180.0°]$. This stands in sharp contrast to the low-angle Gender–Science solution and is too large a shift to be explained by numerical regularization alone. The same estimation procedure that recovers partial contextual overlap in the Gender–Science IAT recovers near-orthogonal geometry in the independent task, confirming that the angle tracks task context.

On the independent task, the full interference model outperforms the quadratic baseline in leak-free hold-out RMSE (5.230 vs. 5.607), while the best-performing model is the cosine-only boundary case (RMSE 4.804). This shows that angular geometry carries predictive leverage when the task occupies a different contextual regime, and that the full sine-plus-cosine form is not always necessary near the orthogonal boundary.

Block-dependent phase structure

Block-specific pooled phase parameters were estimated to examine whether the inferred interference angle reflects a stationary property or an aggregate over heterogeneous contexts. The resulting structure exhibits systematic variation. Block 3 is associated with a lower

pooled angle (MAP = 21.00°), whereas blocks 4, 6, and 7 cluster at higher values (MAPs ranging from 26.25° to 29.00°). This indicates that contextual overlap is not constant across the IAT and that the aggregate interference angle recovered at the population level represents a mixture over block-specific geometries rather than a single stationary parameter.

Blockwise quadratic coefficients, by contrast, point in nearly the same direction in the (d_1, d_2) plane (coefficient angles 145°–148°) and differ mainly in norm. The polynomial model can absorb block-dependent variation only by refitting nuisance coefficients within each block; those coefficients do not map onto a single interpretable geometric dimension. The angle θ places different blocks on a common contextual-rotation scale—low-dimensional structure that local RMSE does not reward.

Limits of individual-level interpretation

Participant-specific interference angles can be estimated, but they are weakly identified under sparse binning. The recovery analyses show that low true angles are especially vulnerable to upward bias and poor interval coverage when fewer than roughly 10,000 participants are aggregated. In the present design, θ is therefore best interpreted as a pooled descriptor of contextual overlap at the population level rather than as a stable individual-difference parameter. The population-level identification is valid and robust: the bootstrap analysis shows that $\theta \approx 17°$–$18°$ is consistently recovered from any large random subsample ($N \geq$ 30,000) of the dataset.

Reproducibility

All analyses were performed using deterministic grid profiling and closed-form regression rather than stochastic posterior sampling. Scripts and configuration files are provided with the project materials so that all reported results can be regenerated.